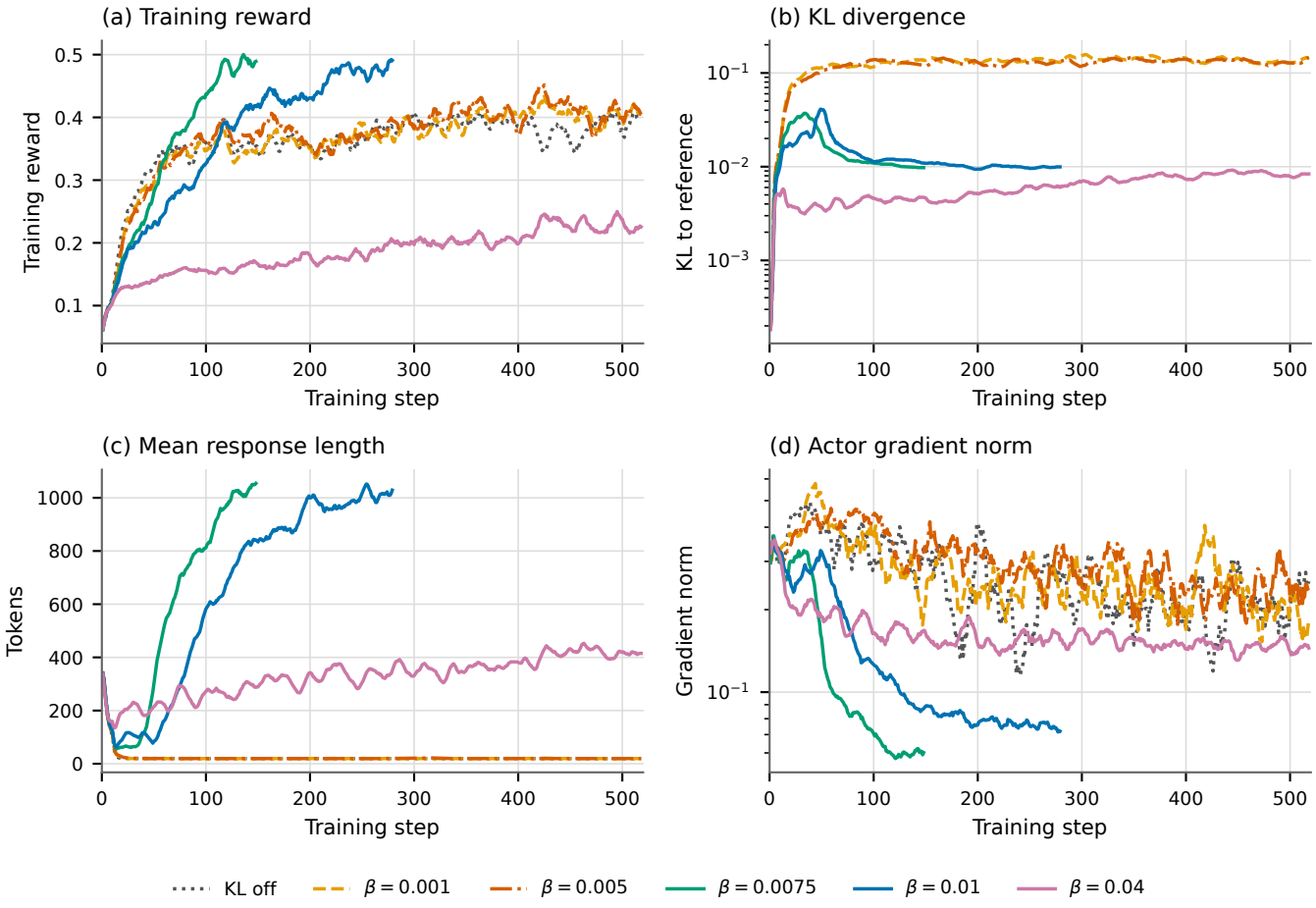

(a) Training reward
Training reward
0.5
0.4
0.3
0.2
0.1
0
100
200
300
400
500
Training step
(b) KL divergence
KL to reference
10^{-1}
10^{-2}
10^{-3}
0
100
200
300
400
500
Training step
(c) Mean response length
Tokens
1000
800
600
400
200
0
0
100
200
300
400
500
Training step
(d) Actor gradient norm
Gradient norm
10^{-1}
0
100
200
300
400
500
Training step
KL off
$\beta = 0.001$
$\beta = 0.005$
$\beta = 0.0075$
$\beta = 0.01$
$\beta = 0.04$